Abiral Shrestha

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Education

Gettysburg College, Gettysburg, PA May 2028
Bachelor of Science in Computer Science, Minor in Mathematics GPA: 3.7

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Discrete Mathematics, Linear Algebra, Databases, Computer Security
- **Honors:** Dean's List (Fall 2024–Present), Presidential Scholar, Dwight D. Eisenhower/Conrad N. Hilton Scholarship
- **Certification:** Machine Learning Foundations, Cornell University

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C#, C, R
Frameworks & Libraries: React, Next.js, Express.js, FastAPI, Redis, BullMQ, Socket.io, Redux
Testing & Tools: Jest, Postman, Git, Jupyter Notebook, Figma
Cloud & DevOps: AWS (EC2, S3, CloudFront, SNS, Lambda), Docker, GitHub Actions, Linux
AI/ML: LLMs, Prompt Engineering, LangChain, LightGBM, XGBoost, SHAP, MCP

Experience

Amazon Web Services Aug 2026 – Present
Break Through Tech AI Studio Fellow Remote

- Collaborating with AWS and a cross-functional team to develop a climate-risk forecasting system using **178K+** spatiotemporal observations to predict high-impact sea surface temperature anomalies.
- Building a modular Python pipeline for data ingestion, preprocessing, validation, missing-data handling, feature engineering, and reproducible testing across temporal train, validation, and test datasets.
- Evaluating model quality and reliability using Macro F1, per-class metrics, and SHAP analysis to identify failure patterns and explain model behavior.

Neighborhood Behavioral Science Jul 2026 – Aug 2026
Gettysburg College Center for Career Engagement – Remote Project Participant (Data Science) Remote

- Rebuilt an R-based natural language processing pipeline from Putka et al. (2023) to estimate job-analysis ratings from O*NET occupational text.
- Migrated the pipeline from O*NET 22.0 to 30.3, adapting occupation records, variables, data structures, and file formats while preserving compatibility with the existing analysis.

Swaya (formerly Sponsy) Mar 2026 – May 2026
Software Engineer Intern Remote

- Built core brand-facing workflows in Next.js and REST APIs as part of a 4-person engineering team, supporting onboarding, profile management, and partnership applications for a platform connecting brands with college organizations.
- Designed the PostgreSQL relational schema for users, organizations, applications, and partnership records, establishing the backend data model for core marketplace workflows.

Projects

Cortex - Real-Time Multiplayer Quiz Generator | Socket.io, Redis, BullMQ, LangChain

- Built a real-time multiplayer quiz platform that converts uploaded notes and documents into AI-generated quizzes, letting users create lobbies, join games, and compete through synchronized rounds.
- Designed synchronous and asynchronous quiz-generation workflows using BullMQ, Redis, and inflight locks, keeping multiplayer lobbies responsive while background jobs execute independently.
- Implemented concurrency-safe scoring, round management, schema validation, retries, and failure handling using Redis Lua scripts, delayed jobs, SET NX locks, and PostgreSQL persistence.

Graph Analysis Automation | FastAPI, TypeScript, MySQL, Gemini API **Schneider Prize Winner**

- Built a full-stack FTIR grease analysis platform for **10+** MRG Labs researchers, automating baseline comparison across **1,000+** CSV spectra and generating real-time graphs for oxidation detection.
- Developed FastAPI and Python data-processing pipelines handling **100,000+** spectral data points in under 5 seconds, reducing API latency by ∼**70%** through caching and query optimization.
- Integrated Gemini APIs into modular analysis services with statistical preprocessing and fallback error handling to maintain application availability during external API failures.

Campus Ride Sharing App | TypeScript, Express.js, PostgreSQL, Redis

- Built a full-stack ride-sharing application with TypeScript and real-time functionality, including ride discovery, booking updates, messaging, and scheduling.
- Developed a geospatial ride-ranking algorithm using Haversine distance and an 8-signal weighted scoring model incorporating pickup proximity, departure time, price, and rider preferences.
- Implemented Redis and BullMQ workflows for real-time communication and asynchronous processing while adding validation to prevent duplicate recurring rides.